INCH-POUND
MIL-DTL-24135C(SH)
24 September 2020
SUPERSEDING
MIL-H-24135B(SH)
3 September 1993

DETAIL SPECIFICATION

HOSE, SYNTHETIC RUBBER, WIRE REINFORCED, FOR FLEXIBLE HOSE ASSEMBLIES, GENERAL SPECIFICATION FOR

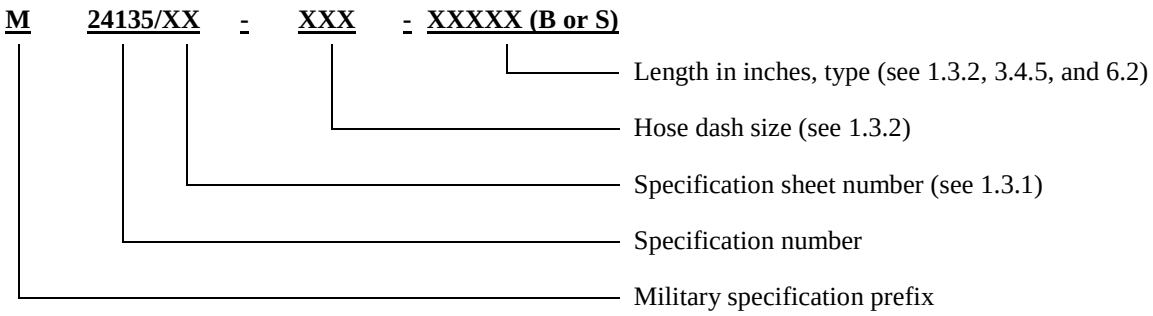
This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for wire reinforced, synthetic rubber hose and hose assemblies for use with reusable end fittings for medium and high pressure service in water, oil, and gas systems. These hoses are not for use in drinking (potable) water systems (see 3.3).

1.2 Classification. Hoses are of the sizes and constructions specified in the individual specification sheets (see 6.2).

1.3 Part or identifying number (PIN) for bulk or straight length hose. The PIN for a bulk or straight (B or S) length hose is a definitive part number that is formed as follows:



PIN example: M24135/12-10X-1200XB describes a MIL-DTL-24135/12 bulk hose with a -10 or $^{5}/_{8}$-inch internal diameter (ID) that is 1200 inches (100 feet) long.

1.3.1 Specification sheet number. Specification sheet numbers that only contain one digit will enter "0" as the first digit. For example, specification sheet 1 will be recorded as "01".

1.3.2 Hose dash size. Hose dash sizes that only contain one digit will have "X" as the second and third digit. When the hose dash size contains two digits, the two digits will be entered followed by an "X" as the last digit. For example, a hose dash size of -40 will be recorded as "40X". For hose dash sizes that contain three digits, all digits will be entered in the PIN. For example, a hose dash size of -128 will be recorded as "128".

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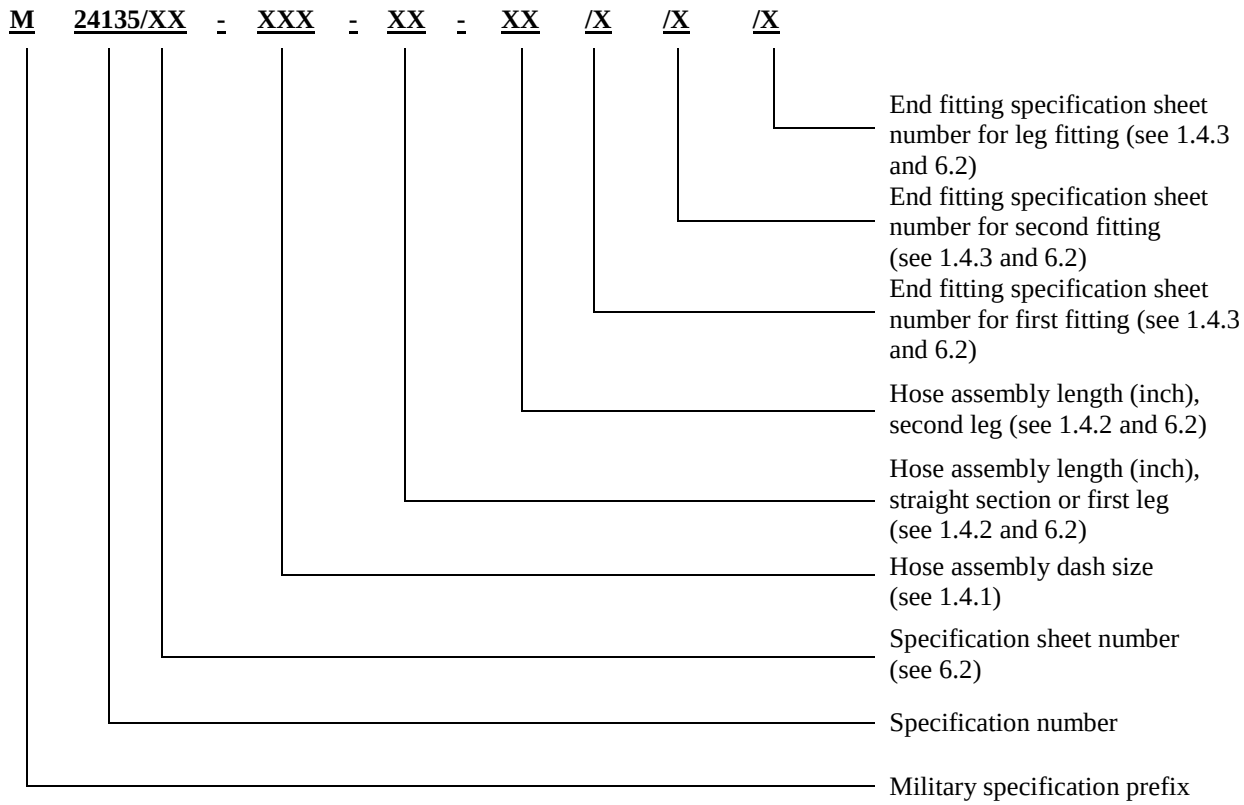
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1.3.3 Hose length type. Hose length type will be identified by either the letter B (bulk length shipped by coil) or S (straight lengths). When the hose length contains unused digits, the length digits are entered followed by an "X" for the remaining digit.

1.4 PIN for hose assemblies. The PIN for a hose assembly is a definitive part number that is formed as follows:



PIN example: M24135/12-10X-36-36/4/4/5 describes a MIL-DTL-24135/12 hose with a -10 or $^{5}/_{8}$-inch ID, 36-inch long first leg, 36-inch long second leg, and a MIL-DTL-24787/4 end fitting for the first and second fitting and a MIL-DTL-24787/5 leg fitting.

1.4.1 Hose assembly dash size. Hose assembly dash sizes, found in table I of the specification sheets, that only contain one digit will have "X" as the second and third digit. For example, a hose dash size of -4 will be recorded as "4XX". When the hose dash size contains two digits, the two digits will be entered followed by an "X" as the last digit. For example, a hose dash size of -40 will be recorded as "40X". For hose dash sizes that contain three digits, all digits will be entered in the PIN. For example, a hose dash size of -128 will be recorded as "128".

1.4.2 Hose assembly length. When the hose assembly is for a straight length of hose, the value for the second leg will be entered as "X". When a hose assembly requires two separate legs of hose, such as for type D (90-degree dogleg), each hose length will be entered.

1.4.3 End fitting specification sheet number. End fittings for MIL-DTL-24135 hoses are in accordance with MIL-DTL-24787. For PIN purposes, only the applicable MIL-DTL-24787 specification sheet number is identified (see 6.2). If a leg fitting is required, the applicable MIL-DTL-24787 specification sheet is identified (e.g., /5 for a 90-degree dogleg). If a leg fitting is not required, the value is identified as "NA". The first and second fittings are for a fitting connecting the hose to a system (e.g., pipe, tube, piping system component). The leg fitting is for connecting a hose to a hose, such as with a 90-degree dogleg configuration.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

(See ASSIST database for list of specification sheets.)

MIL-PRF-2104	-	Lubricating Oil, Internal Combustion Engine, Combat/Tactical Service
MIL-PRF-17331	-	Lubricating Oil, Steam Turbine and Gear, Moderate Service
MIL-H-19457	-	Hydraulic Fluid, Fire-Resistant, Non-neurotoxic
MIL-DTL-24787	-	Fittings, End, Reusable, for Flexible Hose Assemblies, General Specification for
MIL-DTL-32353	-	Hydraulic and Lubricating Oil, Synthetic Hydrocarbon Base
MIL-PRF-83282	-	Hydraulic Fluid, Fire Resistant, Synthetic Hydrocarbon Base, Metric, NATO Code Number H-537

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-177	-	Rubber Products, Terms for Visible Defects of

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM D380	-	Standard Test Methods for Rubber Hose
ASTM D413	-	Standard Test Methods for Rubber Property – Adhesion to Flexible Substrate
ASTM D1141	-	Standard Practice for the Preparation of Substitute Ocean Water
ASTM E162	-	Standard Test Method for Surface Flammability of Materials Using a Radiant Heat Energy Source

(Copies of these documents are available online at www.astm.org.)

SAE INTERNATIONAL

SAE J1942	-	Hose and Hose Assemblies for Marine Applications

(Copies of this document are available online at www.sae.org.)

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2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

3.2 Qualification. Hose furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3).

3.3 Material. Hose materials governed by this specification have not been demonstrated nor are they required to meet the requirements of NSF/ANSI 61; therefore, they shall not be used for potable water systems. Materials shall conform to the requirements in 3.3.1.1 through 3.3.4.

3.3.1 Hose. Hose shall consist of synthetic rubber tube, wire reinforcement, and a synthetic rubber cover.

3.3.1.1 Rubber. The rubber compounds used shall be in accordance with the applicable specification sheet.

3.3.1.2 Reinforcement. Wire used for wire-braid or spiral-wrap reinforcement shall be high-strength carbon steel.

3.3.1.3 Frictioned fabric layer. The frictioned fabric layer, if used, between the tube and reinforcement shall be a cotton or synthetic fiber of uniform texture or a combination of both. The tube shall be fabricated from synthetic rubber compound, as specified on the applicable specification sheet.

3.3.2 Cover. The cover shall be made of a compounded synthetic rubber in accordance with the applicable specification sheet.

3.3.3 End fittings. Hose assemblies' end fittings shall be made of materials in accordance with MIL-DTL-24787 and the applicable MIL-DTL-24787 specification sheet.

3.3.4 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4 Design and construction.

3.4.1 Hose. The hose shall consist of a tube, wire reinforcement, and a cover in accordance with the applicable specification sheet. The hose shall be constructed to meet the requirements of this specification and applicable specification sheet and to retain qualified end fittings in accordance with MIL-DTL-24787 without slippage or leakage when tested as specified herein.

3.4.1.1 Tube. The tube shall have a smooth bore and shall be free from pitting, cuts, dirt, foreign material, mandrel lubricants, or other defects that would affect the performance of the hose.

3.4.1.2 Reinforcement. The reinforcement shall consist of high-strength carbon steel wire-braid or wire spiral-wrap as specified in accordance with the applicable specification sheet. There shall be no broken reinforcing wires nor shall any reinforcing wires be omitted in any length of hose. A frictioned fabric layer may be applied over the reinforcement.

3.4.1.3 Cover. The cover shall be made of synthetic rubber compounded to give maximum resistance to oxidation, weathering, abrasion, and light aging and to meet the requirements of this specification and the applicable specification sheet.

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3.4.1.4 Cover color. Hose covers intended for petroleum-based oils shall be black. Hose covers intended for multi-based lubricants shall be blue.

3.4.2 Concentricity. The maximum and minimum wall thickness shall be obtained from a series of measurements taken around the circumference of the hose but concentrated in the area visually suspected of being thick or thin. Hose wall thickness shall be measured in accordance with ASTM D380. Measurements shall also be taken approximately ½ inch from the hose ends and measured to the nearest 0.001 inch. Concentricity, based upon a full indicator reading between the inside bore of the hose and the outside surface of both the hose and the reinforcement, shall not exceed the values specified in table I.

TABLE I. Concentricity measurements.

Size	Hose ID to OD (inch)	Reinforcement ID to OD (inch)
To -4 (¼-inch)	0.030	0.021
Over -4 to -12 (¾-inch)	0.040	0.028
Over -12 to -32 (2-inch)	0.050	0.038
Over -32	0.062	0.045

3.4.3 Dimensions and pressures. Hose dimensions and pressures shall be in accordance with the applicable specification sheet.

3.4.4 Temperature range. The hose for use with either petroleum-based or phosphate ester-based lubricants shall comply with specified requirements through a temperature range of -40 to +200 °F for all applications. Hose for use with both petroleum-based and phosphate ester-based lubricants (see specification sheets MIL-DTL-24135/12 and /13) shall have a maximum temperature of +300 °F.

3.4.5 Lengths. Hose length shall be the actual length measured along the cover.

3.4.5.1 Hose sizes less than -20 (1¼ inches). Unless otherwise specified (see 6.2), a minimum of 65 percent total lengths furnished shall be 45 feet or longer. A maximum of 10 percent shall be furnished in lengths from 3 to 24 feet.

3.4.5.2 Hose size equal to or greater than -20 (1¼ inches). Unless otherwise specified (see 6.2), 85 percent of total lengths furnished shall be in straight lengths of 12 feet or longer with the remainder in lengths not less than 6 feet.

3.4.6 Hose assemblies. When specified (see 6.2), hose assemblies shall be supplied with hose conforming to this specification, as well as the applicable specification sheet, end fittings in accordance with MIL-DTL-24787, and length(s) of hose leg(s) (i.e., free or actual hose lengths).

3.5 Physical requirements.

3.5.1 Proof pressure. The hose shall show no evidence of leakage, rupture, or deformation when subjected to the proof pressure test specified in 4.5.1.

3.5.2 Stability (dimensional change). When tested in accordance with 4.5.2, hose shall not change in length by more than +2/-4 percent.

3.5.3 Aging. The hose assembly shall show no evidence of aging, such as cracking, separation of plies, or deformation in the form of bulging or collapsing when subjected to the tests specified in 4.5.3.

3.5.4 Oil circulation. There shall be no evidence of leakage through the hose or hose end fitting when subjected to the oil test specified in 4.5.4.

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3.5.5 Impulse. The hose shall show no evidence of failure or leakage when subjected to the impulse test specified in 4.5.5.

3.5.6 Burst pressure. The hose shall show no indication of failure or leakage at less than the specified burst pressure when subjected to the burst test specified in 4.5.6.

3.5.7 Vacuum. There shall be no evidence of ply separation, collapse, or other degradation of the hose when vacuum tested as specified in 4.5.7.

3.5.7.1 Internal support devices. When required for test purposes (see 6.2), internal support devices shall be fitted to the hose assembly to prevent collapse under vacuum service. Support devices shall be a flat wound spring wire strip for hoses up to size -64 (4 inches) and a bonded rubber-ring support cylinder for hoses size -64 and larger.

3.5.8 Cold temperature flexibility. The hose shall show no evidence of cracking or failure when subjected to the test specified in 4.5.8. Cold temperature flexing of hose larger than size -32 (2 inches) will be waived in lieu of a certified affidavit that rubber compounds meet the requirements of ASTM D380.

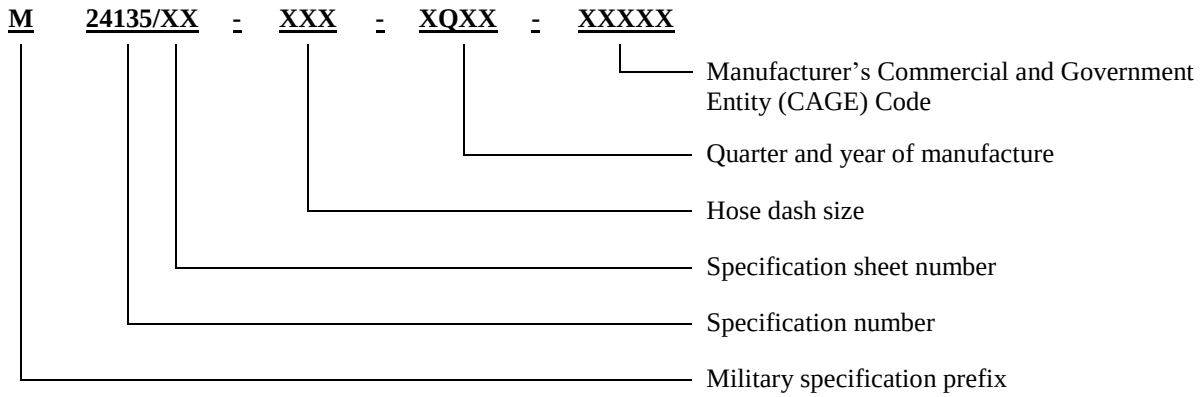
3.5.9 Adhesion between plies. The adhesion between the tube, cover, and reinforcement shall be tested in accordance with 4.5.9 without causing a separation of more than 1 inch.

3.5.10 Fire resistance. The hose shall show no evidence of failure or leakage when subjected to the fire resistance test in accordance with 4.5.10.

3.5.11 Flame spread. Flame spread testing results shall be in accordance with 4.5.11.

3.6 Identification marking.

3.6.1 Hose layline. The hose shall have a permanent and visible layline, either embossed or inlaid, along its entire length. The numerals and lettering shall be not less than 3⁄16 inch high, and each set of markings shall be repeated at intervals not exceeding 20 inches. The layline shall contain, but not be limited to, the following:



Layline example: M24135/12-010-4Q19-53711 represents a MIL-DTL-24135/12 hose, -10 (5⁄8-inch ID) hose dash size, made the fourth quarter of 2019 by NAVSEA.

3.7 Workmanship. The hose shall be uniform in quality and material. The hose tube shall be smooth with no reinforcement visible and shall be free from dirt. The cover shall completely cover the reinforcement plies and shall be free of voids, blisters, cuts, nicks, and other defects that could impair the function and serviceability of the hose in its intended use.

3.8 Age at time of shipment. The hose shall be manufactured not more than 8 calendar quarters (i.e., 2 years) prior to the date of delivery. The calendar quarter of manufacture shall not be counted.

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 Qualification inspection. Qualification inspection shall include the examination of 4.4 and the tests of 4.5.1 through 4.5.11.

4.2.1 Sampling for qualification. Sampling of hose or hose assemblies and the sequence of tests shall be in accordance with both this specification and the applicable specification sheet.

4.3 Conformance inspection. Conformance inspection shall consist of the examinations and tests as specified in table II.

TABLE II. Conformance examinations and tests.

Examination/test	Requirement	Test
Visual and dimensional	3.3, 3.4, 3.6, 3.7, 3.8	4.4
Proof pressure	3.5.1	4.5.1
Stability (dimensional change)	3.5.2	4.5.2
Burst pressure	3.5.6	4.5.6

4.3.1 Lots.

4.3.1.1 Hose. For the purpose of conformance inspection and test sampling, a lot of hose is defined as all hose of the same size and type that is produced in one facility using the same production processes and materials being offered for delivery at one time. At the discretion of the contractor and Government inspector, hose in excess of contractual requirements may be offered for conformance inspection provided the hose is identified with a lot number in the layline in addition to other identification requirements in accordance with 3.6.1.

4.3.1.2 Hose assemblies. For the purpose of conformance inspection and test sampling, a lot of hose assemblies is defined as all the hose assemblies of the same size and type that are produced in one facility using the same production processes and materials being offered for delivery at one time. Such lots may have varying end fittings attached.

4.3.1.3 Lot rejection. Any sample having one or more visual or dimensional defect shall be rejected and shall be cause for rejection of the entire lot represented by the sample. Any sample which fails any of the tests specified in 4.5 shall be cause for rejection of the entire lot. Rejection of a lot in either case shall require that corrective action be taken.

4.3.2 Sampling for visual and dimensional examination.

4.3.2.1 Hose. A sample quantity shall be randomly selected from each lot of completed hose in accordance with table III.

TABLE III. Sampling for visual and dimensional examination.

Linear length in lot (feet)	Lot sample (feet)
≤ 50	1
> 50 to 500	10
> 500 to 1,000	15
Over 1,000	15 + 10 for each 500 feet over 1000 feet

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4.3.3 Sampling for conformance tests.

4.3.3.1 Hose. A sample quantity, in 18-inch lengths, shall be randomly selected from each lot of completed hose in accordance with table IV and shall be tested in accordance with 4.5.2 and 4.5.6 as listed in table II.

TABLE IV. Sampling for conformance tests.

Linear feet in lot	Number of 18-inch samples
≤ 50	1
> 50 to 100	2
> 100 to 500	3
> 500 to 1,000	5
Over 1,000	5 + 1 for each 500 feet

4.3.3.2 Hose assemblies. Hose assemblies shall be fabricated from hose that has been previously examined in accordance with 4.3.2.1 and tested in accordance with 4.3.3.1. Hose assemblies shall be examined to the dimensional requirements specified (see 6.2.2) and proof tested in accordance with 4.5.1. Hose end fittings used in fabrication of hose assemblies shall be qualified in accordance with MIL-DTL-24787.

4.4 Hose Examination. Each of the hose samples selected in accordance with 4.3.2.1 shall be dimensionally examined to determine conformance with the applicable specification sheet and visually inspected for substandard quality conditions and defects in accordance with table II. Terms for visible defects shall be as defined in MIL-STD-177. If one or more visual or dimensional defects are found in any sample, the sample hose shall be rejected, and the entire lot represented by the sample shall be rejected. One hundred percent of the lot may be screened for the defective characteristic(s) or a new lot may be provided which shall be inspected in accordance with table II.

4.5 Tests. Unless otherwise specified (see 6.2), test conditions shall be as specified herein or in the applicable specification sheet.

4.5.1 Proof pressure. Water, oil, or hydraulic fluid in accordance with MIL-PRF-2104, MIL-PRF-17331, MIL-DTL-32353, MIL-PRF-83282, or MIL-H-19457 (applicable to MIL-DTL-24135/12 and /13 only) may be used for proof pressure testing of the hose in accordance with ASTM D380, except that the samples shall be in accordance with 4.2.1 and 4.3.1 and the pressure shall be increased at the following rates:

a. For hose sizes -32 and smaller, the rate of pressure increase shall be not less than 15,000 pounds per square inch (psi) and not more than 25,000 psi per minute.

b. For hose greater than size -32, the pressure increase shall not exceed 1,000 psi per minute.

The pressure rise shall be at a uniform rate. The proof pressure shall be 200 percent of the maximum working pressure specified in the applicable specification sheet, and shall be held for not less than 1 minute nor greater than 5 minutes. Evidence of leakage, rupture, or deformation of the hose shall constitute failure of this test. The passing hose used for this test shall be used for the stability test (see 4.5.2).

4.5.2 Stability (dimensional change). The hose passing the proof pressure test shall be used. The hose samples shall be measured in accordance with ASTM D380 except that they shall be marked at least ½ inch from each end fitting and the distance between the marks measured. The maximum working pressure specified in the applicable specification sheet shall then be applied and the distance between the marks measured at that pressure.

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4.5.3 Aging of samples.

4.5.3.1 Aging of samples for specification sheets /1 through /10. Where required, the assembly shall be completely filled and pressurized to working pressure with the oil in accordance with MIL-PRF-2104, MIL-PRF-17331, MIL-DTL-32353, or MIL-PRF-83282 and held at this pressure while immersed in a non-pressurized, closed-type container or a reflux-type condenser to prevent distillation of the volatile matter in the fluid. Each assembly shall be aged by soaking it in the fluid specified in the applicable specification sheet. Unless otherwise specified in the applicable specification sheet, each assembly shall be aged by being subjected to a temperature of 160 °F+2/-4 percent, first in air, then immersed in synthetic seawater that meets or exceeds the requirements of ASTM D1141 for the times specified below:

a. 76 hours ±¼ hour in air

b. 46 hours ±¼ hour in synthetic seawater

Provisions shall be made to bleed off the excess pressure caused by thermal expansion of the pressurized fluid.

4.5.3.2 Aging of samples for detail specifications /12 and /13. The sample assemblies shall be filled with oil in accordance with MIL-PRF-2104, MIL-PRF-17331, MIL-DTL-32353, MIL-PRF-83282, or MIL-H-19457 (applicable to MIL-DTL-24135/12 and /13 only) and placed in an oven at a temperature of 300±5 °F for 168 hours. During aging, all air must be excluded from the hose bore and provisions made to prevent pressure build-up due to expansion of the oil. At the completion of the 168-hour period, the assembly shall be permitted to cool to room temperature and the oil removed. The assemblies shall be examined for a collapsed or excessively bulged inner tube. Evidence of collapse of the inner tube or excessive bulging of the inner tube at the nipple ends shall constitute failure of the test. Upon completion of the oil aging test, the samples shall be uncapped and placed in an oven at 300±5 °F for a period of 168 hours. At the completion of the 168-hour period, the assembly shall be permitted to cool to room temperature. The assembly shall then be examined for evidence of inner tube cracks and collapse. Evidence of inner tube cracks or collapse shall constitute failure of this test.

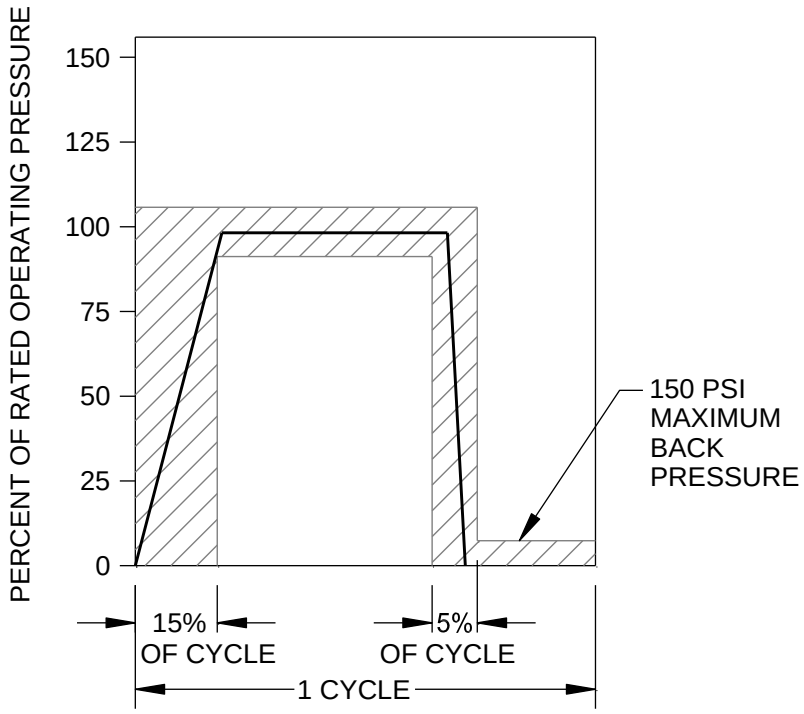
4.5.4 Oil circulation.

4.5.4.1 Ambient temperature circulation. MIL-DTL-24135/1 through /10 sample hose assemblies shall be filled with oil in accordance with MIL-PRF-2104, MIL-PRF-17331, MIL-DTL-32353, MIL-PRF-83282. Sample hose assemblies shall then be pressurized from not less than 75 and not greater than 100 psi. The fluid temperature shall be raised from ambient temperature to 180±10 °F and the fluid circulated through the hose assembly at a flow rate of not less than 3 gallons per minute. The ambient temperature shall be maintained at 75±15 °F. Every 24 hours the pressure shall be increased to maximum working pressure, as listed in the applicable specification sheet, and maintained for not less than 5 minutes. The flow shall be continued for not less than 200 hours. At the completion of this test, the hose assembly shall be pressurized to the maximum working pressure specified in the applicable specification sheet, with the pressure held for not less than 5 minutes.

4.5.4.2 Hot oil circulation. MIL-DTL-24135/12 and /13 sample hose assemblies shall be tested as indicated on the applicable specification sheet.

4.5.5 Impulse. Hose assemblies shall be subjected to impulse testing to the peak pressure specified in the applicable specification sheet. Hose assemblies shall be subjected to an impulse rate from 30 to 80 cycles per minute for the number of cycles stated on the applicable specification sheet while the oil is at a temperature of not less than 120 °F nor greater than 130 °F for the number of cycles stated on the applicable specification sheet. Hose assemblies in sizes -4 through -12 shall be bent through 180 degrees to the minimum bend radius specified in the applicable specification sheet during the impulse test, with both ends rigidly supported. Hose assemblies larger than size -12 shall be dynamically impulsed while straight and one end of the hose shall be free to move during impulsing. Test fluid may flow through the test sample in order to maintain temperature as long as connections do not hamper free movement of one end of the hose. At the beginning of each pressure cycle, the peak pressure specified on the applicable specification sheet shall be attained. The peak pressure shall be held for the pressure time cycle for 100 percent and 125 percent pressure time curves (see figure 1 or 2) as applicable. The applicable working (service) pressure shall be attained before the pressure time curve levels off (see figure 3). Impulse testing for MIL-DTL-24135/12 and /13 hose assemblies shall be as indicated in the applicable specification sheet.

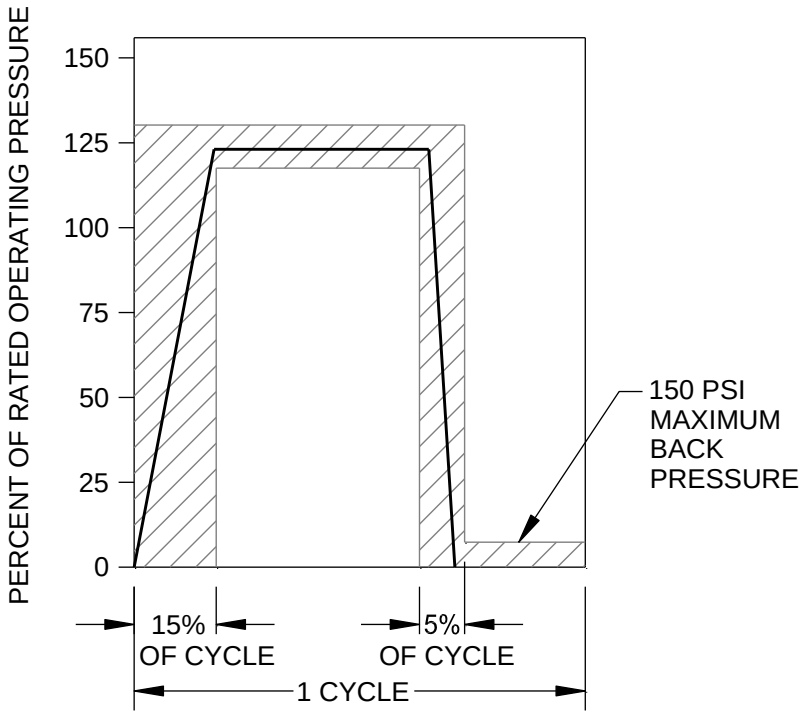
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NOTE: The curve shown on the figure is the approximate pressure time cycle for the pressure impulse to be used in performing the impulse test specified in 4.5.5. The actual pressure time cycle obtained when performing this test should fall within the shaded area.

FIGURE 1. Pressure time curve for impulse test (100 percent square wave).

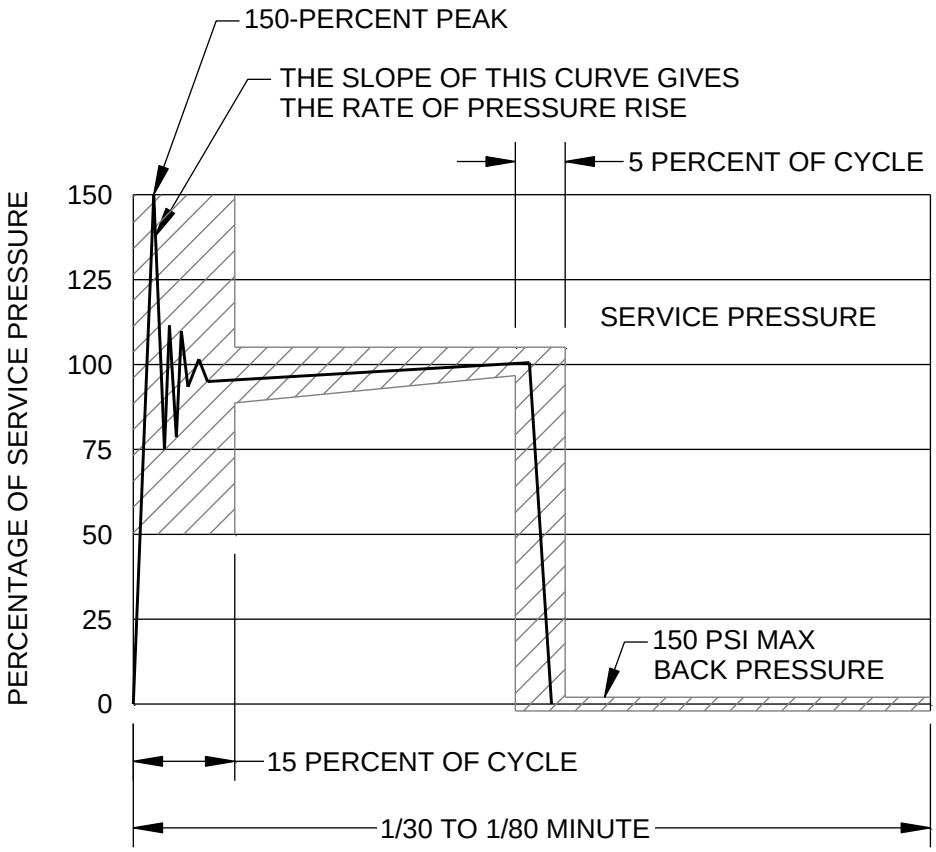
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NOTE: The curve shown on the figure is the approximate pressure time cycle for the pressure impulse to be used in performing the impulse test specified in 4.5.5. The actual pressure time cycle obtained when performing this test should fall within the shaded area.

FIGURE 2. Pressure time curve for impulse test (125 percent square wave).

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NOTE: The curve shown on the figure is the approximate pressure time cycle for the pressure impulses to be used in performing the impulse test specified in 4.5.5. The actual pressure time cycle obtained when performing this test should fall within the shaded area.

FIGURE 3. Pressure time curve for impulse test (150 percent peak wave).

4.5.5.1 Retest and averaging impulse. In the event of a hose failure during impulse testing, an additional hose assembly shall be tested, and the results of all assemblies of the same size tested shall be averaged. The conditions for testing an additional hose assembly shall be as follows:

a. The failed hose shall have reached not less than 95 percent of the required number of impulse cycles.

b. For the purpose of averaging the number of cycles, the maximum number of cycles that can be used from any single tested hose assembly is 110 percent of the required number of cycles, or the actual number of cycles at the time of failure if failure occurs between 100 and 110 percent of the required number of cycles.

4.5.6 Burst pressure. The burst pressure test of the hose shall be conducted in accordance with ASTM D380, except that the samples shall be in accordance with 4.2.1 and 4.3.1. The pressure shall be increased at a uniform rate as follows:

a. For hose size -32 and smaller, the rate of pressure rise shall be not less than 15,000 psi nor greater than 25,000 psi per minute.

b. For hose greater than size -32, the rate of pressure rise shall not exceed 1,000 psi per minute.

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The burst pressure shall be 400 percent of the maximum working pressure specified in the applicable specification sheet. Evidence of leakage, rupture, or deformation of the hose at or below test pressure shall constitute failure of this test. Hoses used for the burst pressure test shall not be acceptable for further use.

4.5.7 Vacuum. The designated samples shall be tested under a vacuum pressure not less than 28 inches of mercury for not less than 30 seconds. If the hose size dictates an internal support, the internal support shall conform to 3.5.7.1.

4.5.8 Cold temperature flexibility. The designated sample hose assemblies shall be cold-temperature tested while filled with fluid in accordance with the applicable specification sheet and enclosed in an environmental chamber at -40± 2 °F for not less than 24 hours. At the end of this period and while at this temperature, the hose assemblies sized -4 through -16 shall be flexed through not less than 180 degrees to the minimum bend radius specified in the applicable specification sheet within 12 seconds. For size -16 and larger, the flexing shall be through not less than 90 degrees to the minimum bend radius specified in the applicable specification sheet within 12 seconds.

4.5.9 Adhesion. Adhesion of the tube and cover to the reinforcement, and plies of reinforcement to each other, shall be determined in accordance with ASTM D413, except that the required minimum weight shall be supported by the test specimen for 3 minutes. Three specimens of each interface shall be tested.

4.5.10 Fire resistance. Fire resistance testing, of any of the qualifying sizes, shall be conducted in accordance with the test method identified by SAE J1942, 30-Minute Fire Test. After the test specimen has returned to ambient temperature, it shall be subjected to a proof pressure test with water of twice the rated working pressure of the hose. Proof pressure shall be held for a minimum of 2 minutes. No external leakage is allowed.

4.5.11 Flame spread. Flame spread testing shall be accomplished in accordance with the method described in ASTM E162. The hose shall be split and flattened to create a test specimen of 6 inches by 18 inches and may be secured to the specimen holder with wire mesh as described in the test specimens section of ASTM E162.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be in accordance with the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Hoses are intended for use in piping systems as flexible connections to resiliently mounted (sound mounted) equipment that may be used for other applications, which do not exceed the pressure or temperature limits of the hose selected, and have an established service life as documented in S6430-AE-TED-010, Piping Devices, Flexible Hose Assemblies. Hose and hose assemblies procured to this specification are form, fit, and function replacements for prior revisions. Existing stock of hose and hose assemblies manufactured to MIL-H-24135B, dated 3 September 1993, are acceptable for use until depleted.

6.2 Acquisition requirements.

6.2.1 Bulk hose or straight hose. Acquisition documents should specify the following:

a. Title, number, and date of this specification and specification sheet.

b. Hose classification, size, and construction (see 1.2).

c. Length of hose required if other than specified (see 3.4.5).

d. Test conditions if other than specified (see 4.5)

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e. Packaging and marking requirements (see 5.1 and 6.6).

6.2.2 Hose assemblies. Acquisition documents should specify the following:

a. Title, number, and date of this specification and specification sheet.

b. Hose classification, size, and construction (see 1.2).

c. Length (free or actual) of hose or hose legs and free length of hose (see 3.4.6).

d. Internal support for vacuum application (see 3.5.7.1).

e. Test conditions if other than specified (see 4.5).

f. Packaging and marking requirements (see 5.1 and 6.6).

g. Type(s) of end fittings (see MIL-DTL-24787 and specification sheets).

h. End fitting PIN for first fitting (pipe end and hose end) (see MIL-DTL-24787 and specification sheets).

i. End fitting PIN for second fitting (pipe end and hose end) (see MIL-DTL-24787 and specification sheets).

j. End fitting PIN for leg fitting (hose end to hose end) (see MIL-DTL-24787 and specification sheets).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No. 24135 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.4 Supersession data. This specification is superseded in part by MIL-DTL-24787, which covers end fittings formerly included in this specification.

6.5 Hose end fitting applicability. When hose end fittings are required for the purpose of hose qualification under this specification for a contract or purchase order for hose assemblies, they must be hose end fittings qualified to MIL-DTL-24787.

6.6 Marking. Marking should be provided on interior packages and exterior shipping containers with the following information (see 6.2):

HOSE TYPE:

DEFENSE SPEC NO: MIL-DTL-24135

DATE OF MFG:

LOT NO:

6.7 Subject term (key word) listing.

Air

Lubricating oil

Nitrogen

Petroleum

Phosphate ester

Piping

Spiral-wrap

System piping

Wire-braid

Water

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6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Preparing activity:
Navy – SH
(Project 4720-2019-007)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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